

# Getting Busy at Home: Remote Work and Fertility in the United States

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## 1 Abstract

Did the spread of remote work change US fertility? Using American Community Survey data for 2018–2024, I instrument local remote work rates with a shift-share design that combines national occupation-specific growth in remote work with each area’s pre-pandemic occupational composition. Among adults aged 18–45, a one-percentage-point increase in the local remote work rate raised the probability of a first birth by 0.024 pp (1.4% of baseline) and lowered the probability of a higher-order birth by 0.065 pp (2.4%). Among couples, the first-birth increase scales with remote-work take-up across couples grouped by whether each spouse’s job is teleworkable. The higher-order decline, however, appears in all couple types and is largest when neither spouse’s job is teleworkable. Preliminary evidence points to housing costs: exposure raised local rents and home values, and the decline is concentrated among renters.

## 2 Introduction

Remote work may provide significant additional flexibility, including by reducing commute times, making work require less supervision, and allowing for alternative schedules and multitasking. Lockdowns following the COVID-19 pandemic resulted in a rapid spread of remote work, and while the push to telework has slightly abated from its high-water mark, more than a fifth of people at work still telework in a given week (BLS, 2024). Bratsberg and Walther (2025) test whether the time freed up by this new flexibility may reduce the cost of taking care of young children and thereby raise birth rates. The authors find that in Norway, pandemic lockdowns raised births by 10%, with the effect concentrated among women in low-flexibility occupations.

While flexibility from remote work can increase the desire for children, remote work overall has an ambiguous effect on births. Having work and home life occur in the same space may have unintended consequences. When examining the adoption of paid paternity leave in Spain, Farré and González (2019) find not only that the reform resulted in fathers providing more childcare at home, but that it also led couples to delay having subsequent children, with fathers reporting a decrease in desired fertility. The authors suggest that this was due to fathers becoming more aware of the costs of childcare. Remote work may similarly decrease the desire of fathers for additional children if it results in a more equal spread of childcare responsibilities.

In this paper I ask whether the spread of remote work changed birth rates in the US and consider alternative mechanisms. I find that increases in the local remote work rate raised first births and lowered higher-order births. While the increase in first births followed workers’ own ability to work remotely, the decline in higher-order births did not. Instead, it was largest among couples in which neither spouse could work from home. One possibility is that switching to remote work for someone with a job that traditionally cannot be done from home is a different type of treatment, a signal of job insecurity, that dominates the additional flexibility that remote work provides to someone with a job that can easily be done remotely. An alternative hypothesis is that the instrument affects the local birth rate in ways unrelated to the adoption of remote work. Given the differential effects on first births compared to higher-order births, I consider whether the instrument affects local housing prices in a way that raises the cost of the space an additional child requires.

### 3 Identification Method and Data

Isolating the causal effect of remote work on fertility poses multiple endogeneity challenges. The choice to work remotely and the choice to start a family are tied to one another and are often made at the same time, while a birth can itself prompt a parent to switch to remote work. A shift-share instrumental variable strategy allows me to identify the effect of local remote work exposure by using an aggregated exogenous shock that is assumed to affect births only through its effect on remote work.

Starting with cross-sectional microdata from the American Community Survey (Ruggles et al., 2025) from 2018-2024, I transform the data into a repeated cross section at the IPUMS consistent PUMA (CPUMA)  $\times$  year level. The endogenous treatment  $X_{it}$  is the share of persons aged 18-45 in location  $i$  who worked remotely in the previous week, while the outcome  $Y_{it}$  is the share with an own co-resident child under one. I later split births by order: a birth is first-order if the respondent has no other own co-resident child and higher-order otherwise. I do not restrict  $X_{it}$  or  $Y_{it}$  to the employed. Remote work requires employment, so if the shock moved people out of work, conditioning on employment would condition on a post-treatment outcome. The shift-share instrument  $z_{it}$  is the inner product of a shift and a share component. For CPUMA  $i$  and year  $t$ ,

$$z_{it} = \sum_k s_{ik} g_{k,t-1}, \quad s_{ik} = \frac{M_{ik}}{\sum_k M_{ik}}, \quad g_{kt} = \frac{\omega_{kt}}{M_{kt}} - \frac{\omega_{k0}}{M_{k0}}, \quad (1)$$

where  $s_{ik}$  is occupation  $k$ 's share of baseline (2018-2019) employment in location  $i$ , and  $g_{kt}$  is the percentage point change in occupation  $k$ 's national remote work rate relative to its base period value.  $M_{ik}$  is the total number of people in occupation  $k$  in location  $i$ , and  $\omega_{kt}$  is the number of people working remotely in occupation  $k$  in year  $t$ . Because it requires a valid occupation, only the instrument is built on employed persons. I lag the shift by one year to better align with the period when the child would have been conceived. The instrument  $z_{it}$  can be thought of as the remote work rate a location would have had if each occupation within it increased its remote workers by the same number of percentage points as that occupation did nationally. This alternative rate excludes the potentially endogenous local variation present in the local remote work rate while retaining exogenous national shocks that can be used for causal identification.

Base-period occupational composition within a CPUMA is plausibly correlated with local birth rates, but the size of an occupation's national shift is assumed to be uncorrelated with unobserved determinants of local birth trends. Since identification in this design rests on the exogeneity of the national shocks, my analysis must cluster standard errors at the occupation level (Borusyak et al., 2022). Borusyak et al. (2022) also show the equivalence between the 2SLS shift-share instrument and the coefficient from a shock-level IV procedure, in which the shock directly serves as the instrument. Therefore, while the shift-share instrument is originally conceived as the sum of the products of the shift and share components, I use the established equivalence to simplify the instrument to the shift component alone (i.e.,  $g_{kt}$ ). This requires reshaping the location-level remote work and birth rates to the occupation-year level while retaining information on each location's remote workers and birth rates. I do this sequentially by first residualizing  $X_{it}$  and  $Y_{it}$  on a set of controls before calculating a mean of each weighted by occupation  $k$ 's baseline employment share in each location.

I then run two-stage least squares estimation, instrumenting the occupation  $\times$  year remote work rate  $\bar{X}_{kt}$  with the lagged percentage point shift in occupation  $k$ 's remote work rate  $g_{k,t-1}$ . The first stage is

$$\bar{X}_{kt} = \pi g_{k,t-1} + \nu_{kt},$$

and the second stage is

$$\bar{Y}_{kt} = \beta \bar{X}_{kt} + \epsilon_{kt}.$$

I confirm that the effect size in the shock-level specification is equivalent to the shift-share approach. Controls include year indicators and each location's baseline remote work share, share of those in tele-

workable occupations (Dingel and Neiman, 2020), foreign-born share, and 2020 state COVID-19 death rate, each interacted with year.

## 4 Results and Discussion

Table 1: Effect of local remote-work exposure on births, by birth order

|                     | (1)                 | (2)                | (3)                 |
|---------------------|---------------------|--------------------|---------------------|
| All births          | −0.0411<br>(0.0089) |                    |                     |
| First births        |                     | 0.0243<br>(0.0028) |                     |
| Higher-order births |                     |                    | −0.0655<br>(0.0064) |
| Total births        | 194,935             | 76,776             | 118,159             |
| Total adults        | 4,508,833           | 4,508,833          | 4,508,833           |
| Clusters            | 529                 | 529                | 529                 |

*Notes:* Shock-level 2SLS (Borusyak et al., 2022). Controls include year indicators and each location’s baseline remote work share, share of those in teleworkable occupations, foreign-born share, and 2020 state COVID-19 death rate, each interacted with year. Occupation-clustered standard errors in parentheses.

Table 1 reports the headline results. A one percentage point increase in the local remote work rate lowers the birth rate by 0.041 percentage points. This net decline masks opposing effects by birth order. First births rise by 0.024 percentage points, or 1.4% of baseline, while higher-order births fall by 0.065 percentage points, or 2.4%. At the average increase in local remote work of 7.45 percentage points, this implies about 0.18 pp more first births and 0.49 pp fewer higher-order births per year.

I next test whether the effect runs through one spouse’s remote work. I match couples (married or cohabiting) and categorize them by whether each spouse’s job can be done from home, following Dingel and Neiman (2020).

Table 2: Effects by couple teleworkability

| Teleworkable job(s):                                       | Both                  | Husband only          | Wife only             | Neither               |
|------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <i>A. Reduced form</i>                                     |                       |                       |                       |                       |
| All births                                                 | −0.00372<br>(0.00071) | −0.00090<br>(0.00083) | −0.00151<br>(0.00070) | −0.00741<br>(0.00066) |
| First births                                               | 0.00307<br>(0.00037)  | 0.00506<br>(0.00047)  | 0.00536<br>(0.00034)  | 0.00196<br>(0.00030)  |
| Higher-order births                                        | −0.00803<br>(0.00070) | −0.00588<br>(0.00099) | −0.00801<br>(0.00066) | −0.01137<br>(0.00075) |
| <i>B. First stage: at least one partner works remotely</i> |                       |                       |                       |                       |
|                                                            | 0.189<br>(0.009)      | 0.166<br>(0.007)      | 0.137<br>(0.006)      | 0.068<br>(0.003)      |

*Notes:* Married and cohabiting couples; each partner’s occupation is classified as teleworkable following Dingel and Neiman (2020). Treatment is the local share of couples in which at least one partner works remotely.

Table 2 reports the results from this analysis, with both the first stage and reduced form coefficients. Take-up of remote work occurs as expected such that the instrument increases the probability of a spouse working from home (first stage) more if both spouses have teleworkable jobs than if just one does and both more than if neither have a traditionally teleworkable occupation (e.g., registered nurse). First

births also rise across all four groups and rise least where neither job is teleworkable. Higher-order births fall in all four group, but the decline is largest where neither spouse has a job that traditionally can be worked from home ( $-0.0114$ , against  $-0.0080$  where both can).

This unexpected finding for higher-order births among couples with non-teleworkable jobs may be the result of a violation of shock exogeneity. One possibility is that remote work raises local housing costs such that households that cannot telework bear the costs of exposure without gaining the benefits of the added flexibility. A different reading is that for workers in non-teleworkable occupations a switch to remote work is itself a sign of economic disruption. If a switch to remote work signals job insecurity, it would likely suppress births in general. However, those couples may also expect the switch to be temporary and have a first birth sooner than they had already planned, which fits the positive first-birth result.

To test the housing mechanism, I examine whether the shock affects gross rents and home values. I also split the sample between renters and homeowners and test for differences between the two in the decline of higher-order births. Preliminary evidence shows that the shock is correlated with higher rents and home prices. Additionally, the decline in higher-order births is almost entirely among renters while first births rise among owners, even though take-up of remote work is nearly the same for the two groups. These results are suggestive that a space constraint may be driving decisions over having an additional child that is being affected by remote work.

To investigate the insecurity mechanism, I split each couple teleworkability type by college degree status. Disaggregating couples where neither job is teleworkable, I find that the first stage effect of the shock on remote work status is highest for couples where both have a college degree and lowest for those without. At the same time the reduced form effect of the shock on births is roughly equivalent across education. The gap between couples with and without degrees leaves the possibility for job insecurity to drive this effect, which I will test next.

## References

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